

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Minutes

QUESTIONS: 30

Time Duration: 30

Total Marks:30

Annexure A

MCQ

Code of Conduct:

I ___ V.SURYA SATVIKA(192524254)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.S.SATVIKA

Annexure A

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment c) An agent that only follows pre-defined scripts
- b) An agent that uses a knowledge base and logical inference to decide actions d) An agent that learns from rewards

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$ c) $\neg Q \rightarrow \neg P$ only
- b) $\neg P \vee Q$ d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions c) A conjunction of literals only
- b) A conjunction of disjunctions (clauses) d) A sequence of implications

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass c) If something passes, it is a student
- b) Every entity that is a student passes d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

6. What is the core function of a logical agent in AI?

- a) Reacting only to current percepts
- b) Using logic and knowledge representation to reason and make decisions
- c) Performing physical actions without internal representation
- d) Relying solely on sensors for navigation

7. Which component of a logical agent is responsible for storing facts and rules about the world? a) Inference Engine

- b) Perception Unit
- c) Knowledge Base (KB)
- d) Action Selector

8. What does a knowledge-based agent do immediately after obtaining a percept from the environment?

- a) Selects an action
- b) Executes the action
- c) Updates the Knowledge Base with the new percept
- d) Reasons to derive new information

9. Which operation is used to add new information to a Knowledge Base?

- a) ASK b) TELL c) PERCEIVE d) UPDATE

10. What is a proposition in formal logic?

- a) A command or request
- b) A declarative statement that is either true or false
- c) A question regarding the environment
- d) A letter representing a variable

11. Which of the following is NOT a proposition?

- a) "The sun rises in the east."

- b) "5 + 3 = 10."
- c) "Close the door."
- d) "Chennai is in India."

12. Which logical operator is represented by the symbol $\neg$?

- a) AND
- b) OR
- c) NOT
- d) Implication

13. The AND ($\wedge$) operator is true only when:

- a) At least one proposition is true
- b) Both propositions are true
- c) Both propositions have different truth values
- d) Neither proposition is true

14. A proposition that is always true, regardless of the truth values of its variables, is called a:

- a) Contradiction
- b) Tautology
- c) Contingency
- d) Equivalence

15. In an implication ($P \rightarrow Q$), the statement is false only when:

- a) P is false and Q is true
- b) P is true and Q is false
- c) Both P and Q are false
- d) Both P and Q are true

16. What is the number of rows in a truth table for a logical expression with 3 propositions? a) 4 b) 6 c) 8 d) 9

17. Which of these represents one of De Morgan's Laws?

- a) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- b) $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$
- c) $\neg(\neg P) \equiv P$
- d) $P \vee \neg P \equiv \text{True}$

18. A proposition that is sometimes true and sometimes false is known as a:

- a) Tautology
- b) Contradiction
- c) Contingency
- d) Implication

19. Which logical connective is true if and only if both propositions have the same truth value?

- a) Implication ($\rightarrow$)
- b) Biconditional ($\leftrightarrow$)
- c) OR ($\vee$)
- d) NOT ($\neg$)

20. According to operator precedence, which operator should be evaluated first?

- a) AND ($\wedge$)
- b) OR ($\vee$)
- c) NOT ($\neg$)
- d) Implication ($\rightarrow$)

21.What is the process of deriving a conclusion from given premises called?

- a) Perception b) Representation c) Inference d) Predication

22.Which inference rule follows the form: If P, then Q; P; Therefore, Q?

- a) Modus Tollens b) Modus Ponens c) Hypothetical Syllogism d) Resolution

23.Which valid reasoning pattern is represented by: $P \rightarrow Q$; $\neg Q$; Therefore, $\neg P$?

- a) Modus Ponens b) Disjunctive Syllogism c) Modus Tollens d) Addition

24.What can be concluded from the premises $(P \rightarrow Q)$ and $(Q \rightarrow R)$ using Hypothetical Syllogism?

- a) $P \wedge R$ b) $P \vee R$ c) $P \rightarrow R$ d) $\neg P \rightarrow \neg R$

25.The rule that states if "P or Q" is true and "Not P" is true, then "Q" must be true is called:

- a) Simplification b) Conjunction c) Disjunctive Syllogism d) Resolution

26.Which rule allows you to conclude "P" from the premise "P and Q"?

- a) Addition b) Simplification c) Conjunction d) Modus Ponens A

27.The inference rule Resolution follows which symbolic form?

- a) $P \therefore P \vee Q$ b) $P, Q \therefore P \wedge Q$ c) $P \vee Q, \neg P \vee R \therefore Q \vee R$ d) $P \rightarrow Q, Q \rightarrow R \therefore P \rightarrow R$

28.Which of the following is an example of the "Addition" rule?

- a) P and Q, therefore P
b) P, therefore P or Q
c) P, Q, therefore P and Q
d) P or Q, not P, therefore Q

29."If it rains, the ground is wet. The ground is wet. Therefore, it rained." This is an example of which invalid reasoning pattern?

- a) Denying the Antecedent b) Affirming the Consequent c) Modus Ponens d) Double Negation

30.What is the name of the invalid pattern: $P \rightarrow Q$; $\neg P$; Therefore, $\neg Q$?

a) Affirming the Consequent b) Denying the Antecedent c) Modus Tollens d) Hypothetical Syllogism

CO3-AT2
MCQ's TEST

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- ① [b] ✓
- ② [b][c] ✗
- ③ [b] ✓
- ④ [b] ✓
- ⑤ [b] ✓
- ⑥ [b] ✓
- ⑦ [c] ✓
- ⑧ [c] ✓
- ⑨ [d] ✗
- ⑩ [b] ✓
- ⑪ [c] ✓
- ⑫ [c] ✓
- ⑬ [b] ✓
- ⑭ [b] ✓
- ⑮ [c] ✗
- ⑯ [c] ✓
- ⑰ [a] ✓
- ⑱ [c] ✓
- ⑲ [b] ✓
- ⑳ [c] ✓
- ㉑ [c] ✓
- ㉒ [b] ✓

⑧ ✓
24
—
30

- 23 [c] ✓
- 24 [c] ✓
- 25 [a] ✗
- 26 [b] ✓
- 27 [d] ✗
- 28 [b] ✓
- 29 [b] ✓
- 30 [c] ✗

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions